

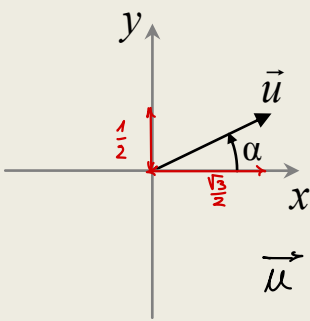
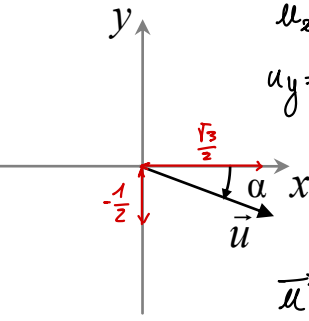
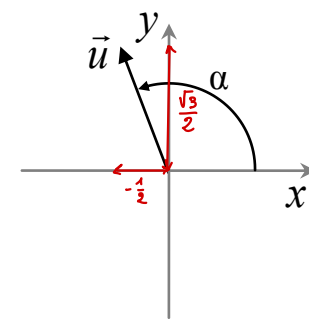
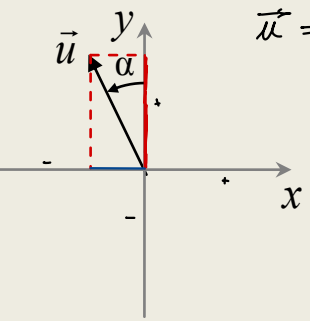
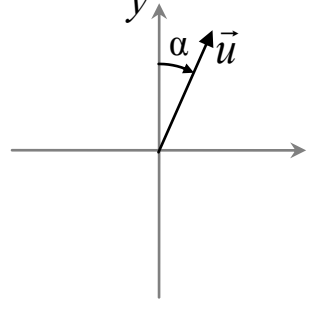
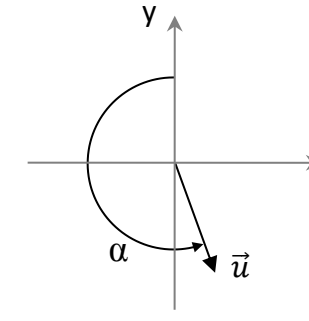
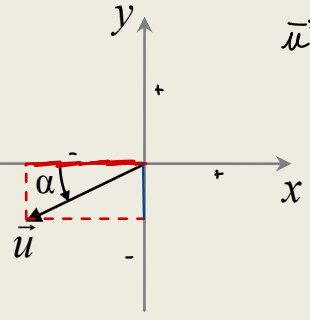
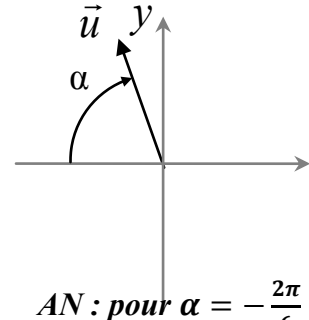
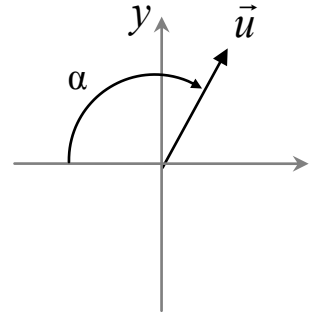
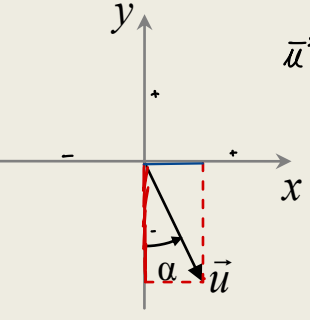
PROJECTION d'un vecteur sur une base orthonormée

Soit $B = (\vec{x}, \vec{y}, \vec{z})$, une base orthonormée directe. ($\vec{z}$ vers nous)
Soit $\vec{u}$ un vecteur unitaire.
La valeur de l'angle α est algébrique.

Ecrire les composantes du vecteur $\vec{u}$ dans la base B , sous la forme $\pm \cos \alpha$ ou $\pm \sin \alpha$.

Conclusion : Méthode pour projeter un vecteur :

On réalise toujours une figure de notation/projection telle que les angles soient positifs et soient inférieurs à $\frac{\pi}{2}$. (Attention aux vecteurs !! il faut que la base soit directe)

 $u_x = \vec{u} \cdot \vec{x} = \ \vec{u}\ _x \ \vec{x}\ _x \cos(\alpha)$ $u_y = \vec{u} \cdot \vec{y} = \ \vec{u}\ _y \ \vec{y}\ _y \cos(\alpha - \frac{\pi}{2}) = \sin(\alpha)$ $\vec{u} = u_x \vec{x} + u_y \vec{y}$ $= \cos(\alpha) \vec{x} + \sin(\alpha) \vec{y}$ AN : pour $\alpha = \frac{\pi}{6}$ $\vec{u} = \frac{\sqrt{3}}{2} \vec{x} + \frac{1}{2} \vec{y}$	 $u_x = \vec{u} \cdot \vec{x} = \ \vec{u}\ _x \ \vec{x}\ _x \cos(\alpha)$ $u_y = \vec{u} \cdot \vec{y} = \ \vec{u}\ _y \ \vec{y}\ _y \cos(\alpha - \frac{\pi}{2}) = \sin(\alpha)$ (angle <u>en haut à droite</u> donc + et pas -) $\vec{u} = \cos(\alpha) \vec{x} + \sin(\alpha) \vec{y}$ AN : pour $\alpha = -\frac{\pi}{6}$ $\vec{u} = \frac{\sqrt{3}}{2} \vec{x} - \frac{1}{2} \vec{y}$	 $u_x = \vec{u} \cdot \vec{x} = \ \vec{u}\ _x \ \vec{x}\ _x \cos(\alpha)$ $u_y = \vec{u} \cdot \vec{y} = \ \vec{u}\ _y \ \vec{y}\ _y \cos(\alpha - \frac{\pi}{2}) = \sin(\alpha)$ (idem) $\vec{u} = \cos(\alpha) \vec{x} + \sin(\alpha) \vec{y}$ AN : pour $\alpha = \frac{4\pi}{6}$ $\vec{u} = -\frac{1}{2} \vec{x} + \frac{\sqrt{3}}{2} \vec{y}$
 $\vec{u} = -\sin(\alpha) \vec{x} + \cos(\alpha) \vec{y}$		
 $\vec{u} = -\cos(\alpha) \vec{x} - \sin(\alpha) \vec{y}$	 AN : pour $\alpha = -\frac{2\pi}{6}$	
 $\vec{u} = \sin(\alpha) \vec{x} - \cos(\alpha) \vec{y}$	